



Lilliputian Robots: The Micro-Bots Revolution Is Here!

Micro-robotics deals with small-scale robots of micrometres size. A typical macro-scale mobile robot is a self-contained, untethered and re-programmable machine that can perceive, move and learn in a given environment to accomplish a given task repeatedly and with precision



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In his book *The Diamond Age*, Neal Stephenson opines that countries and other powerful stakeholders are likely to fight silent wars with each other using a very large number of tiny robots, in the future. However, the real culprits behind these micro wars will remain hidden.

Micro-robots, also known as micro-bots, typically have dimensions ranging from a fraction of a millimetre up to several millimetres, and are able to access less than one-millimetre space directly with minimal invasion and untethered operation. Micro-bots are made of sub-millimetre-scale components, such as micro-actuators, micro-sensors and micro-mechanisms, and are constructed using micro-fabrication methods, which are different from conventional macro-scale machining techniques. Their locomotion and physical interaction in a given environment are dominated by micro-scale physical forces and effects. Volume-based forces such as inertial forces, gravity and buoyancy become comparable to surface area and perimeter-based forces such as viscous forces, drag, friction, surface tension and adhesion.

Micro-bots can access hard-to-reach places and offer the promise of extending human presence to locations that are lethal or filthy, thus offering an edge over normal robots. These are inexpensive, unobtrusive and find much wider usage than larger robots.

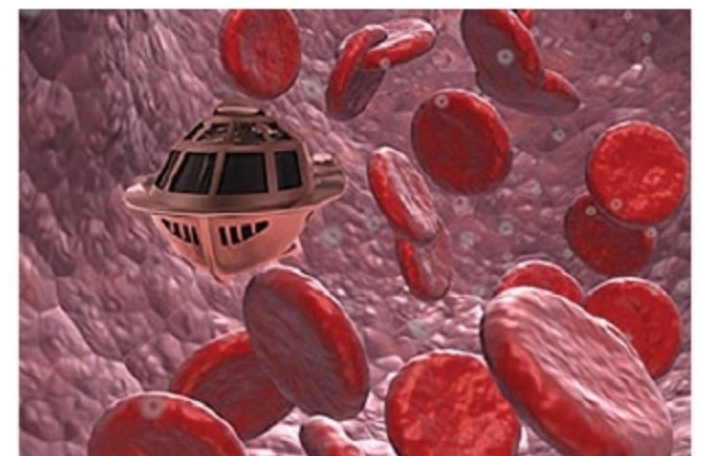
A micro-bot can be either autonomous or insect-like. An autonomous micro-bot, like any larger robot, includes its own onboard computer to control and pilot

the machine, and operates independently. An insect-like micro-bot operates in a fleet of identical units controlled by a central computer. Insect-like robots usually work in large groups as swarms and almost behave like ants in an anthill or bees in a hive.

How micro-bots developed

The famous quote by physicist Richard Feynman in 1959, "There is plenty of room at the bottom," which triggered the developments in the fields of nanotechnology and bio-nanotechnology, is treated as the birth of micro-system technology (MST). The first ideas of micro-robotics appeared with the emergence of an active investigation of MST, and rapid development of micro-fabrication technologies made it possible to provide micro-components for micro-bots at a reasonable cost.

Integrated circuits (ICs) discovered in 1949 and electronic microscopes helped micro-scale fabrication. With time, miniaturisation technology enabled building smaller and smaller devices as well as enhancing



A scene from the 1966 sci-fi movie *Fantastic Voyage*, which popularised micro-bots (Credit: <https://vtropes.org>)